

Metal Affinity Protein Precipitation: Effects of Mixing, Protein Concentration, and Modifiers on Protein Fractionation

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Received November 22, 1994/Accepted June 15, 1995

Previous work by us and others has shown that mixing impacts apparent protein solubility in single protein precipitations. In this work, we probe the effects of contacting conditions on fractional precipitation behavior at the bench scale. We have chosen metal affinity precipitation as our model system; the kinetics of this mode of precipitation are very rapid and largely irreversible and, consequently, mixing conditions govern the extent of fractionation and purity of the product in such a process. Our experimental strategy involved a three-pronged approach to control the effects of contacting conditions on precipitate yield, purity, and particle size distribution. First, we studied the impact of process variables that control precipitant concentrations in the reactor including impeller speed and precipitant addition rate. Second, we controlled the rate of precipitation by changing the initial protein concentration to alter the protein-protein collision rate. Third, we examined the role of the molecular-level kinetics of affinity precipitation by using modifiers that compete with surface moieties to bind the metal ion, thereby reducing its availability. Our model process and protein system consisted of zinc precipitations of mixtures of bovine serum albumin and bovine γ -globulins, carried out at a nominal 1-L scale; glycine was examined as a modifier. Faster impeller speeds and lower precipitant addition rates increased the desired protein yields, decreased purities, and reduced average precipitate particle size. Higher initial protein concentrations were found to produce precipitates with higher yields, lower purities and diminished particle sizes. Experiments with glycine indicated that modifiers in the precipitant solution serve to increase product purity, decrease yield, and increase the average particle size in bench-scale precipitations. © 1995 John Wiley & Sons, Inc. Key words: plasma fractionation • mixing • protein precipitation • metal affinity

INTRODUCTION AND LITERATURE SURVEY

Precipitation processes have been employed extensively in industrial protein purification operations. They are used to reduce processing volumes, to separate protein from cell debris, to fractionate protein mixtures via differential precipitation, and to solidify product protein in finishing op-

erations.^{1,3,27,28,38} Research done previously, and our recent work in particular, has shown that rapid, kinetically irreversible phenomena in salt-induced precipitations result in apparent protein solubilities that are considerably impacted by reactor contacting conditions.^{14,22} This suggests that mixing conditions could also play a critical role in determining fractionation performance in multiprotein systems. In this work, we attempt to elucidate the impact of mixing on fractionation behavior in a two-protein, group-specific affinity precipitation system.

Our model process and system consists of the semi-batch differential precipitation of mixtures of bovine serum albumin (bSA) and bovine γ -globulins (b γ Gs) induced by zinc acetate at a nominal 1-L scale. The system is a mimic for classic plasma fractionation which has been studied extensively.^{10,24,32,34} In the early 1950s, several researchers published a considerable amount of work advocating precipitation of plasma proteins with zinc ions and they postulated that groups on the protein surface were involved in binding zinc ions.^{10–12,8,9,16,20} It was suggested that the groups involved in the binding were predominately surface-exposed histidines and, to a lesser extent, cysteines and tryptophans.¹⁷

The metal ions selectively precipitate proteins from solution by coordinating the lone pair electrons of heteroatoms on the sidechains of these amino acids. This binding reduces the solubility of the protein by changing its isoelectric point.^{13,35,39} However, the majority of the precipitation at low protein concentrations is thought to occur due to the interprotein crosslinking between metal ion-bound and free surface groups.⁴¹ Protons in solution also compete with the zinc ions for binding sites; during the course of a metal affinity precipitation there is usually a change in pH as protons are displaced from their coordination sites by the stronger binding zinc ions. Precipitations using zinc are very rapid and are found to be kinetically irreversible.^{25,32} This results in considerable mixing control of the yields, purities, and particle sizes in these processes.^{22,32}

More recently, affinity precipitations of serum proteins were effected by Maeda and co-workers^{23,25} who studied the extent of precipitation of human fibrinogen using zinc

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chloride and copper chloride. They examined fibrinogen solubility by measuring supernatant turbidity at 350 nm and showed that the rate of redissolution of the precipitates was extremely low, rendering precipitation essentially irreversible. Zaworski and Gill⁴² used zinc chloride to rapidly precipitate and concentrate various recombinant proteins such as porcine urokinase, tissue-type plasminogen activator analog FK2P, and human interleukin-1 β (IL-1 β) from cell culture media. They observed that the precipitated proteins could be resolubilized by decomplexing with ethylene diamine tetraacetic acid (EDTA) solutions and that 100% activity was recovered.

Van Dam and co-workers⁴¹ have studied metal affinity precipitation of human hemoglobin and sperm whale myoglobin using both bis-copper chelates and copper sulfate. They found that the distribution and steric accessibility of histidines on a protein surface are crucial in determining the efficiency of precipitation. In proteins in which the histidines were fewer in number and sterically hindered, the use of copper ions chelated with either ethylene glycol bis(β -aminoethyl ether)-*N,N'*-tetraacetic acid or polyethylene glycol di-iminodiacetic acid was found to improve precipitation performance relative to copper sulfate by alleviating some of the steric constraints.

In our initial investigation of the effect of precipitation kinetics on the precipitation of bSA and b γ Gs with zinc acetate as the precipitant, we estimated the apparent Smoluchowski time constants for aggregation via stopped-flow/rapid mixing turbidimetry.²⁹ To assess the effect of the extent of mixedness on precipitate yield, purity, and particle size in zinc-induced precipitation, we employed a three-pronged strategy.

First, we reduced the local concentration of the precipitant in the reactor by increasing the intensity of mixing. We can control the intensity of mixing via the impeller speed and precipitant addition rates. In doing this, the concentration of unbound zinc is reduced throughout the reactor, and as a consequence, less overprecipitation results.^{22,32} Second, we varied the initial protein concentration to control the rate of precipitation; simple collision theory dictates faster crosslinking rates at higher protein concentrations. Third, we changed the effective binding kinetics of zinc by adding competing chelating species or "modifier" to the precipitant mixture,³⁹ we used glycine which can readily complex the zinc ions. Complexation is somewhat different from merely reducing the concentration of zinc in solution; the presence of this complex enables much better control of the rates of aggregation in the system by acting as a zinc ion buffer. This also results in a more uniform distribution of the precipitating agent in a precipitation reactor.

For each of these approaches, we determined the yield and purity of the precipitate phase; we assumed that the b γ Gs were our desired product. We also considered the final precipitate particle size distribution as a function of these process parameters as reactor operating conditions that maximize particle size and density enable more effective solid-liquid separations downstream.

MATERIALS AND METHODS

Materials

The bovine serum albumin (lot 13H0328) and bovine γ -globulins (lot 112H9335), admixed as a plasma protein mimic in our zinc affinity precipitation experiments, were obtained from Sigma Chemical Co. (St. Louis, MO). bSA has a molecular weight of 66.5 kDa, and a pI of 4.9, and a primary sequence with 17 histidines.^{31,36} b γ Gs have an average molecular weight of 168 kDa, a pI range of 4.5 to 7.5, and approximately 22 histidines on average.³⁶ The number of histidine residues that are surface exposed in either bSA or b γ Gs have yet to be determined. In human blood plasma, albumin constitutes about 60% of the total protein content with concentrations of 40.4 g/L; γ -globulins constitute about 11% of the total protein content with concentrations of 7.4 g/L.³⁷

Zinc acetate (lot 102H0169), zinc oxide (lot 120H0504), glycine (lot 53H2618), and buffer salts used to make the metal-affinity precipitant solutions were also obtained from Sigma. For experiments done in the absence of glycine, zinc acetate precipitant solutions of the required molarity were made up in 100 mM Tris buffer, pH 7.2. For experiments done in the presence of glycine, precipitant solutions were prepared from the appropriate mixtures of ZnO and glycine in 100 mM Tris buffer, pH 7.2, using the method of Surgenor and co-workers.³⁹ It should be noted that Tris can also form a weak complex with metal ions such as Zn⁺⁺.²

Determination of Apparent Aggregation Time Constants

The kinetic time constants for the aggregation of bSA and b γ G under different initial protein and modifier concentrations were measured by monitoring changes in turbidity upon precipitation using a rapid mixing setup in a Lambda-6 UV-VIS spectrophotometer (Perkin-Elmer, Norwalk, CT). The rapid mixing device consisted of a standard quartz cuvette in which a short length of Tygon tubing connected to a syringe was introduced. This system was operated manually with protein solution charged to the cuvette and a precipitant bolus added via the syringe. Mixing times were rapid in this system; an acid-neutralization reaction of phenolphthalein monitored at 530 nm was essentially complete within 1 second.

Due to the manual nature of the device operation, all rapid mixing experiments were performed in duplicate. The data acquisition was done by interfacing the spectrophotometer with an IBM PS/2 computer; absorbance at 350 nm was monitored after mixing the appropriate protein and precipitant solutions. The absorbance at wavelength λ , $A(\lambda)$, is related to the weight-averaged molecular weight, $\bar{M}_w$, via²⁹

$$A(\lambda) = \frac{16\pi K(\lambda)C_o\bar{M}_w}{6.909M_1} \quad (1)$$

where $K(\lambda)$ is a wavelength-dependent coefficient, C_0 is the initial concentration of monomeric protein, and M_1 is its molecular weight.

If a purely collisional precipitation mechanism is assumed, Smoluchowski's population balance equation can be used to describe the precipitation kinetics. A further assumption of an invariant aggregation rate constant admits an analytical solution of Smoluchowski's equation for the protein aggregate weight-averaged molecular weight as a function of time.¹⁹

$$\bar{M}_w(t) = \bar{M}_w(0) + 2k_{app}C_0t \quad (2)$$

where k_{app} is the apparent Smoluchowski aggregation rate constant. An estimate for the characteristic time for aggregation, τ_{app} , which is inversely proportional to k_{app} , can be fitted to the linear region of the raw turbidity trajectories by combining Eqs. (1) and (2) to yield

$$\tau_{app} = \frac{M_1}{2k_{app}C_0} = 7.275C_0K(350 \text{ nm}) \left[\frac{dA(350 \text{ nm})}{dt} \right]_{\text{linear}}^{-1} \quad (3)$$

To determine the values of $K(350 \text{ nm})$ for bSA and b γ G, their weight-averaged molecular weights in low concentration solutions were assumed to be approximately equal to their monomeric molecular weights.^{18,30} $K(350 \text{ nm})$ for bSA and b γ Gs were estimated from a linear least squares fit of concentration versus absorbance data as $6.427 \times 10^{-3} \text{ mL/mg}$ and $7.96 \times 10^{-3} \text{ mL/mg}$, respectively. Concentrations of protein solutions used to determine $K(350 \text{ nm})$ were in the range of 0 to 13 mg/mL. The region of the turbidity trajectories used to estimate the time constant was obtained by choosing a contiguous subset of the data points in the rapidly increasing portion of the curve that minimized the 95% confidence intervals of the fitted slope. The non-negative linear least squares slope fitting and confidence interval minimization routines were programmed in MATLAB 4.0 (MathWorks, Natick, MA).

Precipitation Reactor and Conditions

Precipitations were done in a semibatch mode. We used a 1-L jacketed reactor (Kontes, Vineland, NJ) thermostatted with a recirculating water bath set at a temperature of 25°C and fitted with baffles. A three-blade marine impeller obtained from Lightnin (Model TSM2510, Avon, NY) was employed to mix the reactor contents. The empty reactor was initially charged with 400 mL of protein solution. A computerized peristaltic pump (Model 7550-60, Cole-Parmer, Chicago, IL) was then used to deliver 400 mL of precipitant to this protein solution at a preset addition rate. Thus, both the initial protein and precipitant solutions are diluted by a factor of two on the macroscopic scale by mixing alone. A diagram of the reactor setup used in these experiments can be found elsewhere.²²

Micro-mixing and macro-mixing times in the reactor under good and poor mixing conditions were estimated pre-

viously by determining the selectivity of a series-parallel dye reaction in conjunction with a mixing model.²² *Good* mixing was defined as reactor contacting conditions which resulted in quick homogenization of the liquid fed to the reactor, that is, a high impeller speed (540 rpm) and low precipitant addition rate (8 mL/min). The mixing time constants determined were $\tau_{micro} = 10.2 \text{ s}$ and $\tau_{macro} = 10.2 \text{ s}$, where τ_{micro} and τ_{macro} refer to micro- and macro-mixing time constants.²² *Poor* mixing was said to exist when the impeller was operated at low speed (60 rpm) and the precipitant was added rapidly (200 mL/min). The corresponding mixing time constants were $\tau_{micro} = 966.0 \text{ s}$, $\tau_{macro} = 102.0 \text{ s}$.²² In the context of protein precipitation, the "goodness of mixing" refers to the extent and longevity of zones of nonequilibrium precipitant concentrations.

Experimental Protocol

Preliminary experiments were done to assess the impact of mixing on the extent of precipitation of each protein individually. Both bSA and b γ G were precipitated at the 1-L scale at initial concentrations of 2 mg/mL using 30 mM zinc acetate as the precipitant. The only variable in these experiments was the extent of mixing in the reactor, delineated as good and poor. Particle size and solubility were measured at the end of each experiment; the experiments were done in duplicate to obtain estimates for the standard deviation of our results.

A large number of system parameters impact the outcome of our more extensive two-protein precipitation experiments. To address the effects of the most important of these parameters efficiently, we used an experimental design to plan our experiments and to obtain statistically meaningful results.²² We varied three parameters in our experimental design: initial protein concentrations with values of 1 and 2 mg/mL; the extent of mixing using good and poor mixing as defined above; and initial glycine concentration with values of 0, 30, and 60 mM.

Measurement of bSA and b γ G Yield and Purity

We used size exclusion chromatography (SEC) to separate mixtures of bSA and b γ G and to estimate their concentrations in the supernatants following precipitations. A Protein-Pak 300SW (7.5 mm $\times$ 13 cm) size exclusion column was obtained from Waters (Bedford, MA). This column had a molecular weight range of 10 to 400 kDa for globular proteins. The mobile phase for the analysis was 100 mM phosphate buffer, pH 6.5, with 100 mM NaCl at a flow rate of 0.5 mL/min.

Using the supernatant solubility measurements, we estimated the purity and yield of b γ G in the precipitate phase; in our system, we assumed the γ -globulins to be the desired protein due to their highly diminished solubility in the presence of zinc ions. *Purity* was defined as the fraction of desired protein present in the precipitate phase:

Purity, $P \equiv$

$$\frac{\text{mass of desired protein (b}\gamma\text{G) precipitated}}{\text{total mass of protein precipitated}} \times 100\%$$

or

$P =$

$$\frac{[C_{\text{b}\gamma\text{G}}V]_{\text{initial}} - [C_{\text{b}\gamma\text{G}}V]_{\text{final}}}{[(C_{\text{b}\gamma\text{G}} + C_{\text{bSA}})V]_{\text{initial}} - [(C_{\text{b}\gamma\text{G}} + C_{\text{bSA}})V]_{\text{final}}} \times 100\% \quad (4)$$

where C_i is the concentration of the species i in the supernatant, and V is the fluid volume in our semibatch reactor system. Because the volume of the solid precipitate was found to be negligible in comparison to the total volume of the supernatant, we approximated V_{final} by the sum of the initial volumes of the protein and precipitant solutions. *Yield* was defined as the percentage of the total amount of desired protein that was obtained in the precipitate phase:

Yield, $Y \equiv$

$$\frac{\text{mass of desired protein (b}\gamma\text{G) precipitated}}{\text{initial mass of desired protein (b}\gamma\text{G)}} \times 100\%$$

or

$$Y = \frac{[C_{\text{b}\gamma\text{G}}V]_{\text{initial}} - [C_{\text{b}\gamma\text{G}}V]_{\text{total}}}{[C_{\text{b}\gamma\text{G}}V]_{\text{initial}}} \times 100\% \quad (5)$$

Based on the above definitions, the ideal fractionation experiment would give us a yield and purity of 100%.

Measurement of Particle Size Distributions

A Microtrac-FRA 9200 series particle size analyzer (Leeds and Northrop Corp., North Wales, PA) was used to measure the precipitate particle size distributions. Samples were collected at the end of each precipitation experiment for particle size analysis and were added to the analyzer sample well with the aid of a Pasteur pipette. There was some mechanical disruption of the aggregates due to the flow in the sample well, but the effect was small (see Fig. 1). We expect significant disaggregation to occur only in the presence of a metal chelator.

The analyzer's resident software was used to deconvolve the light scattering data, giving the volume percent particles as a function of particle diameter; the analyzer uses Mie scattering theory⁵ for particle sizes between 0.13 μm and 1 μm and Fraunhofer diffraction theory⁶ for particle sizes between 1 μm and 700 μm . Fines were defined as the cumulative volume percent of particles below a size of 20 μm in the particle size distribution.

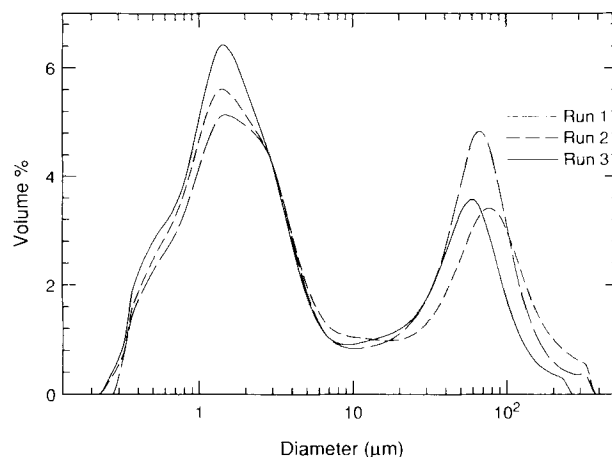


Figure 1. Representative particle size distribution following metal affinity precipitation. Precipitation conditions: b γ G(0) = 2.0 mg/mL, poor mixing, 30 mM zinc as precipitant.

RESULTS AND DISCUSSION

Characteristic Aggregation Times

The apparent Smoluchowski characteristic aggregation times obtained by fitting the stopped-flow turbidity data are shown in Table I. Sample turbidity trajectories for the aggregation of bSA and b γ G are shown in Figure 2. Three important pieces of information can be obtained from the table. First, the aggregation time constants of bSA are much larger than those of b γ G. As mentioned earlier, this is likely due to the greater number of surface-exposed histidines present on the γ -globulins. Second, there is an order-of-magnitude increase in the apparent time constants for aggregation as the modifier concentration increases from 0 to

Table I. Apparent Smoluchowski aggregation time constants estimated from rapid-mixing turbidity measurements of single- and two-protein systems at different initial protein and glycine concentrations.

bSA (mg/mL)	b γ G (mg/mL)	Glycine (mM)	τ_{app} (s)
2	0	0	6.3 $\pm$ 0.4
2	0	30	72.6 $\pm$ 7.0
2	0	60	183.8 $\pm$ 19.5
0	2	0	< 1.0
0	2	30	1.3 $\pm$ 0.1
0	2	60	17.8 $\pm$ 0.4
1	0	0	68.5 $\pm$ 9.8
1	0	30	574.7 $\pm$ 49.8
1	0	60	> 1400
0	1	0	2.6 $\pm$ 0.7
0	1	30	22.2 $\pm$ 2.9
0	1	60	206.1 $\pm$ 25.1
1	1	0	< 1.0
1	1	30	5.8 $\pm$ 0.7
1	1	60	113.0 $\pm$ 25.2

Results reported as the mean $\pm$ SD of duplicate experiments.

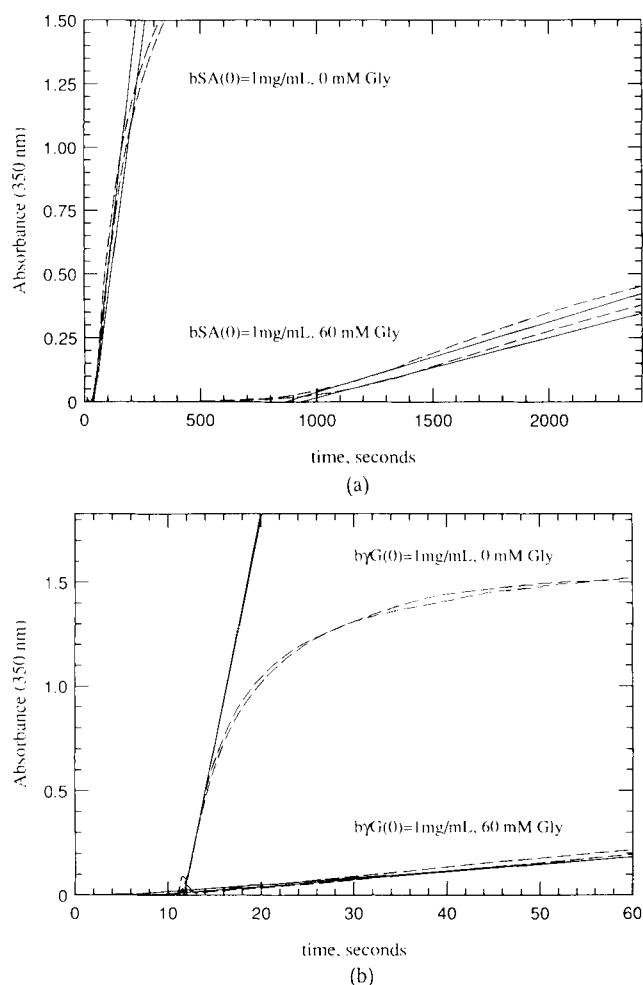


Figure 2. Aggregation turbidity trajectories as a function of modifier concentration. (a) bSA and (b) b γ G. Duplicate measurements are shown. Initial protein concentrations are 1.0 mg/mL, precipitant concentrations are fixed at 30 mM zinc. Solid lines indicate linear fits of turbidity data as indicated by Eq. (3) in the text.

30 to 60 mM. This implies that the aggregation time constants for increased modifier concentrations exceed the micro-mixing time constants and mitigate the impact of mixedness considerably. Third, decreasing protein concentrations tend to increase the aggregation time constants significantly. Therefore, precipitations carried out at higher initial protein concentrations can be expected to be governed, to a greater extent, by mixing in the system. This concentration dependence also indicates that simple collision aggregation kinetics is not an adequate description of the metal affinity precipitation mechanism.

Single Protein Experiments: Effect of Mixing

The solubility of bovine serum albumin was measured under both good and poor mixing conditions. The results, shown in Table II, indicate that 37% of the 0.8 g of bSA initially charged to the reactor precipitated under both poor and good mixing conditions. The final solubility does not

depend very strongly on the extent of mixing in the reactor as predicted by the similarity of τ_{micro} and τ_{app} .

Particle size distribution data were measured as described previously. To facilitate the understanding and interpretation of the complex particle size data, they have been tabulated in terms of the percent of fines in the reactor in Table II. Rapid mixing results in distributions that have a larger fraction of fines. This could be due to two different effects. First, higher impeller speed results in greater shear levels in the reactor, forming fines from attrition of larger precipitate particles. Second, quicker homogenization of the precipitant results in a larger number of nucleation points, and hence, smaller primary aggregate particles.

γ -Globulins were precipitated under both good and poor mixing conditions as well. Table II shows that they precipitate to a much larger extent than the bSA. However, they are less sensitive to mixing conditions than the bSA; this runs contrary to our expectations from the relative magnitudes of τ_{micro} and τ_{app} . The greater extent of precipitation can be explained by the larger number of binding sites available for the zinc ions on the surface of the γ -globulins.

The γ -globulin precipitates are much smaller than the bSA precipitates. This can be clearly seen from the increased fraction of fines that are present in the cumulative particle size distribution shown in Table II. The precipitates generated using this protein also show that they are influenced to a much lesser extent by mixing conditions. This could be due to the differences in the two proteins with respect to the vastly different number of metal-binding sites on each of their surfaces. The greater number of binding sites of the γ -globulin, as compared to bSA, results in more rapid precipitation kinetics and results in a greater number of nucleation points spread throughout the mixed zone in the reactor (see Table I). This causes the precipitates to comprise smaller particles.

Finally, we precipitated both bSA and b γ Gs individually under poor/good mixing conditions.

Poor/good mixing conditions can be described as follows. Initially, precipitant addition accompanied by precip-

Table II. Results of yield and particle size measurements from single protein precipitations under different mixing conditions.

bSA (mg/mL)	b γ G (mg/mL)	Mixing conditions	Yield (%)	Volume % fines (<20 μ m)
2	—	Good	40 $\pm$ 0	53 $\pm$ 3
2	—	Poor	34 $\pm$ 3	54 $\pm$ 38
2	—	Poor/good	—	92 $\pm$ 1
—	2	Good	77 $\pm$ 1	93 $\pm$ 2
—	2	Poor	76 $\pm$ 1	76 $\pm$ 9
—	2	Poor/good	—	75 $\pm$ 1

Mixing conditions: good, impeller speed 540 rpm, precipitant addition rate 8 mL/min; poor, impeller speed 60 rpm, precipitant addition rate 200 mL/min; and poor/good implies salt addition and precipitation done under poor mixing conditions followed by an interval of good or high shear mixing conditions. Initial concentration of precipitant was maintained at 30 mM zinc acetate and 0 mM glycine in all of these experiments. Results reported as mean $\pm$ SD for duplicate experiments.

itation was done under poor mixing conditions. Once all the precipitant was added and equilibrium established, the impeller speed was increased to 540 rpm for an interval of 15 min; this established high shear conditions analogous to good mixing for that period of time. At the end of the experiments particle size distributions were measured.

The aim of this experiment was to determine whether the final particle size was sensitive to the shear in the reactor. Although the interpretation of the data may be complicated by aging effects, the results, shown in Table II, indicate that, for bSA, there is a huge effect of shear on the final precipitate particle size. This could be owing to the larger particles that are generally formed by bSA due to its lower nucleation rate. On the other hand, the particle size of the b γ Gs are less sensitive to the effect of shear after formation of the precipitate phase. This is likely due to the smaller particles that are formed by the rapidly nucleating b γ G precipitate phase. The effect of aging in these experiments is deterministic; an inverse experiment performed under good/poor mixing conditions would likely lead to different results.³³

Two-Protein Experiments: Effect of Mixing

Protein mixtures were precipitated under conditions described earlier. Table III shows yield and purity of the b γ Gs in the precipitate phase as a function of precipitation conditions. At the 1-L scale, our experiments show that poor mixing results in higher yields than good mixing. This was expected since persistence of high concentration precipitant zones results in a greater extent of precipitation. Surprisingly, our experiments also show that poorer mixing resulted in a higher purity of the precipitate phase; poor mixing conditions impact the less soluble protein, b γ G, to a greater extent than the more soluble protein, bSA. *If this result translates well to larger scales, it raises very inter-*

esting possibilities regarding the design of precipitation reactors that are purposely poorly mixed.

Table III shows the impact of the intensity of mixing on the precipitate particle size distributions obtained from our experiments at different mixing intensities. It can be seen that poor mixing conditions result in a larger population of average-size particles. This could be due to reduced shear-driven breakage at the lower impeller speed associated with poor mixing. Higher impeller speeds result in smaller particles due to shear or a more rapid homogenization of the precipitant, or both. Again, quicker homogenization results in the formation of a greater number of smaller nuclei, and fewer numbers of large particles.

In some of the two-protein precipitations there is a greater yield of bSA compared with the single-protein system. This is due to the rapidity of the precipitation. As can be seen from the values of τ_{app} in Table I, the presence of γ -globulins increase the overall rate of precipitation resulting in greater coprecipitation of the bSA. However, it is hard to distinguish between the greater coprecipitation and the accompanying occlusion of unprecipitated bSA resulting in its anomalously lower solubility.

Two-Protein Experiments: Effect of Initial Protein Concentration

Initial protein concentration was found to be a very important parameter in determining precipitate yield, purity, and particle size distributions; that this is the case is not surprising as metal affinity precipitation is a reaction-based phenomenon.

Table III shows the effect of this parameter on purity and yield of b γ G in a precipitation step. At higher initial protein concentrations, the rate of precipitation increases due to the availability of a larger concentration of histidine sites for initial ion binding and crosslinking. As borne out by our

Table III. Results of yield, purity, and particle size measurements from two-protein precipitations under different mixing conditions.

bSA (mg/mL)	b γ G (mg/mL)	Mixing conditions ^a	Glycine (mM)	Yield (%)	Purity (%)	Fines (%)
2	2	Good	0	73 $\pm$ 9	57 $\pm$ 2	66 $\pm$ 48
2	2	Poor	0	82 $\pm$ 8	61 $\pm$ 3	45 $\pm$ 59
1	1	Good	0	68 $\pm$ 8	54 $\pm$ 6	70 $\pm$ 41
1	1	Poor	0	68 $\pm$ 11	71 $\pm$ 1	44 $\pm$ 54
2	2	Good	30	62 $\pm$ 2	69 $\pm$ 2	75 $\pm$ 16
2	2	Poor	30	70 $\pm$ 3	77 $\pm$ 6	18 $\pm$ 18
1	1	Good	30	34 $\pm$ 2	76 $\pm$ 3	61 $\pm$ 37
1	1	Poor	30	56 $\pm$ 3	85 $\pm$ 9	16 $\pm$ 1
2	2	Good	60	53 $\pm$ 2	100 $\pm$ 10	59 $\pm$ 16
2	2	Poor	60	39 $\pm$ 22	95 $\pm$ 8	5 $\pm$ 2
1	1	Good	60	27 $\pm$ 10	100 $\pm$ 10	93 $\pm$ 8
1	1	Poor	60	12 $\pm$ 11	100 $\pm$ 10	11 $\pm$ 10

^aMixing conditions as defined in Table II.

Initial concentration of precipitant was maintained at 30 mM in all of those experiments. Results reported as the mean $\pm$ SD for duplicate experiments.

τ_{app} data in Table I, this would also be equivalent to precipitating a protein at the same concentration with a larger number of histidines on its surface. This increased rate of aggregation causes more of the contaminating protein to coprecipitate with the less soluble protein and a concomitant reduction in purity. However, the increased rate of aggregation also results in a greater extent of precipitation. There is clearly a trade-off between yield and purity.

The influence of initial protein concentration on the final precipitate particle size distribution is also shown in Table III. It can be seen that increasing the concentration of protein results in lower particle sizes. This is consistent with the formation of a larger number of nuclei at higher protein concentrations.³²

Two-Protein Experiments: Effect of Modifier

Precipitations were done in the presence of glycine as a modifying agent. As shown in Table III, the consequent slowing in the kinetics of the precipitation reaction has major implications for the purity and yield of the precipitate. The b γ Gs aggregate at a much slower rate, resulting in a precipitate phase that is considerably purer in the presence of glycine. Consequently, fewer molecules of bSA are precipitated due to diminished scavenging by slowly precipitating b γ G.⁷ The slower rate of aggregation also dictates that there is less overprecipitation.³²

The diminishing rate of aggregation in the presence of modifier also results in fewer nuclei being formed; this results in bigger particle sizes as shown in Table III. The final particle size observed is a balance of two different effects: first, increased shear causes a greater number of collisions; and second, increased shear also causes greater breakage.³² This balance can be shifted by reducing the number of metal-bound species on the protein surface through the addition of a competing modifier, thereby decreasing the efficiency of each collision. Therefore, it can be observed from Table III that, in the absence of modifier, the mixing effects and the high collision efficacy balance each other and there is no net effect of poor mixing conditions and shear. Addition of a modifier may shift the aggregation kinetics toward the reaction limited regime; one can envisage that this would result in denser precipitates with reduced inclusion of mother liquor and lesser solvent-exposed surface area.⁴

CONCLUSIONS

We have used a zinc/glycine precipitant system to conduct an experimental study on the impact of process parameters on the fractional precipitation of plasma proteins. We found that precipitant composition had a much greater effect on purity and yield than the initial protein concentration and mixing conditions precipitation. Increased modifier concentrations in the precipitant make-up were found to increase the purity and decrease the yield of the precipitate phase. Our bench-scale mixing results indicate that reactor config-

urations that engender poorer mixing may be more desirable if the protein of interest has a relatively large number of surface-exposed metal ion binding sites. These conclusions seem to be in direct contradiction to traditional views on choosing optimal fractional precipitation conditions.³² However, we believe that these results reflect the fact that a good understanding of the mechanism of the precipitation reaction is crucial to design precipitation reactor vessels and to specify operating conditions.

On the other hand, metal affinity induced precipitations have very different solubility behaviors for a given protein. Locally high metal ion concentrations in the reactor do not necessarily translate into either faster precipitation rates or lower solubilities. In fact, recent modeling work by us to describe the metal affinity precipitation of proteins using gelation theory shows that locally high metal ion concentrations actually reduce the rate of precipitation as the number of unbound sites decreases.²¹

Arguments similar to this can be extended to pH-induced precipitations of proteins and nucleic acids as high initial proton concentrations in acid feeds need not necessarily result in lower solubilities. This contrasts with traditional salting-out precipitations, in which the Cohn-Setchenow solubility equation dictates that higher local salt concentrations increase the rate of precipitation,²² implying that poorer mixing would cause greater precipitation of protein.

Mixing conditions in reactors were also found to impact the particle size distributions observed in metal affinity precipitations. As seen previously in the literature, strong shear fields in the presence of good or vigorous mixing resulted in considerable breakage of aggregate particles and enhanced production of fines in the reactor,^{3,15,26} such a situation has to be avoided from a process standpoint due to loss of yield in solid-liquid separations downstream of precipitation.⁴⁰

We have been able to measure particle size distributions for single-protein and two-protein precipitations. Single-protein precipitate particle size distributions show that individual protein types influence the final distributions very strongly. Our results indicate that there is an influence of bSA on the size distribution of a predominately b γ G precipitate. Modifiers such as glycine slow down the kinetics of the affinity precipitation reaction by competitively binding the precipitant ions; this potentiates the production of larger and purer precipitate particles due to the formation of fewer nuclei for aggregation. The potential application of modifiers such as glycine will be dictated by product purity and formulation requirements.

Metal affinity precipitations are best used when proteins in mixtures show large differences between each other in the number of surface-exposed histidines. By examining the solubilities of individual components of a mixture with respect to metal ion concentration, the zinc concentration that will maximize the precipitation of the desired component may be selected. If such an optimal concentration of precipitant can be found, the protein concentration could be raised until a limit is reached when increasingly rapid precipitation kinetics would cause unacceptably high co-

precipitation or scavenging of the undesired protein. The optimum protein concentration to be employed for the maximum throughput would depend on the proteins under consideration, the metal ion used, and the extent of mixing in the reactor. Precipitates that are formed can be recovered by using stronger chelating agents such as EDTA. Future work will involve modeling the affinity precipitation mechanism and incorporating it into our mixing model.

We are very grateful to Dr. Fred Rothstein of Bioseparations Consultants, Long Beach, CA, for suggesting this problem to us and for many helpful discussions. We thank Mr. Nick Sopchak and Professor Misiolek of the Department of Materials Engineering at RPI for their help with the particle size analysis equipment. Funding for this work was provided by NSF Grant CTS-9211666.

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